

Osteoarthritis



Introduction

Osteoarthritis is the most common form of arthritis. Some people call it degenerative joint or wear and tear arthritis. It occurs most frequently in hands, hips, and knees. Osteoarthritis occurs most in older people; it affects more than 46 million men and women in the United States.

Definition

Osteoarthritis is a degenerative joint disease that can affect the many tissues of the joint. It is by far the most common form of arthritis.

Pathophysiology

Osteoarthritis is traditionally thought of as a 'wear and tear' disease which occurs as we age. However, recent research suggests otherwise. The pathogenesis of OA involves a degradation of cartilage and remodeling of bone due to an active response of chondrocytes in the articular cartilage and the inflammatory cells in the surrounding tissues and the release of enzymes from these cells break down collagen and proteoglycans, destroying the articular cartilage and the exposure of the underlying subchondral bone results in sclerosis and the joint space is progressively lost over time.

Incidence

Osteoarthritis (OA) is the most common joint disorder in the United States and osteoarthritis is the commonest rheumatic disease in Egypt, with a prevalence of 8.5%. The disease is more common in females than males. Symptomatic knee OA occurs in 10% men and 13% in women aged 60 years or older.

Classifications

Grade 0 (none): definite absence of x-ray changes of osteoarthritis

Grade 1 (doubtful): doubtful joint space narrowing and possible osteophytic lipping

Grade 2 (minimal): definite osteophytes and possible joint space narrowing

Grade 3 (moderate): moderate multiple osteophytes, definite narrowing of joint space and some sclerosis and possible deformity of bone ends

Grade 4 (severe): large osteophytes, marked narrowing of joint space, severe sclerosis and definite deformity of bone ends

Causes

1. Heredity
2. Obesity (overweight lead to inability to move)
3. Lack of activities
4. Sport injuries like knee trauma
5. Wear and tear like joint damage such as torn cartilage, dislocated joints, ligament injuries or joint malformation.

Risk factors

Older age. The risk of osteoarthritis increases with age.

Sex. Women are more likely to develop osteoarthritis, though it isn't clear why.

Obesity. Carrying extra body weight contributes to osteoarthritis in several ways, and the more you weigh, the greater your risk. Increased weight adds stress to weight-bearing joints, such as your hips and knees. Also, fat tissue produces proteins that can cause harmful inflammation in and around your joints.

Joint injuries. Injuries, such as those that occur when playing sports or from an accident, can increase the risk of osteoarthritis.

Repeated stress on the joint. If your job or a sport you play places repetitive stress on a joint, that joint might eventually develop osteoarthritis.

Genetics. Some people inherit a tendency to develop osteoarthritis.

Bone deformities. Some people are born with malformed joints or defective cartilage.

Certain metabolic diseases. These include diabetes and a condition in which your body has too much iron

Some occupational activities such as kneeling, squatting, climbing, heavy lifting, vibration and sales worker and machine worker increase the risk of osteoarthritis

Symptoms

- **Pain**
- **Joint stiffness** a common marker of osteoarthritis is stiffness in the joint. Stiffness may be most noticeable first thing in the morning or after a prolonged period of inactivity (such as sitting in a car or airplane). The stiffness usually resolves within 30 minutes of light activity, which can warm up the joints

- **Loss of range of motion** While stiff joints may be able to warm up and regain their full range of motion, that is not always the case. It is possible for joints to become less flexible.
- **Swelling** : the joint may swell due to excess fluid buildup.(Fever, Red, warmth)
- **Deformity** : Bone spurs or other changes in the joint may lead to a noticeable physical deformity. For example, the arthritic joints in the fingers may become enlarged or the knees may appear bowlegged.
- **Loss of function**

Signs

- Mild to moderate swelling around the joint
- Crepitus in movement
- Limited range of motion
- Pain with movement in joint
- Joint tenderness
- Mild inflammation and warmth over the joint
- Joint deformity and enlargement
- Joint instability
- Bone spurs : Friction and pressure between a joint's bones can lead to bone degeneration and the growth of bone spurs.
- Loss of functioning addition to a loss of range of motion, other changes may make it more difficult for the joint to work the way it used to.

Diagnosis

Physical exam

1. Mild to moderate swelling around the joint
2. Crepitus in movement

4. Pain with movement in joint
5. Joint tenderness
6. Mild inflammation and warmth over the joint
7. Joint deformity and enlargement
8. Joint instability

Lab and test

Currently, there is no standard lab test to verify the presence of osteoarthritis. Lab tests may be used to rule out other problems, such as rheumatoid arthritis, which also causes joint pain. Lab tests may require drawing fluid from the joint (arthrocentesis) or taking a blood sample.

Imaging

1. X-rays are typically used to confirm the diagnosis of osteoarthritis. X-rays can reveal asymmetric joint space narrowing
2. While magnetic resonance imaging (MRI) is a more sensitive imaging method, it is used less often than X-rays due to cost and availability. MRI scans show cartilage, bone, and ligaments.
3. DEXA Scan: the most important type used to measure the percentage of mineral density.

Treatment

Pharmacological treatments

There is no cure for OA, but medication and other therapies that don't involve drugs can help to relieve pain. As a last resort, a damaged joint may be surgically fused or replaced with one made of a combination of metal, plastic and/or ceramic.

Medications

Analgesics. These are pain relievers and include acetaminophen and opioids.

Nonsteroidal anti-inflammatory drugs (NSAIDs). These are the most commonly used drugs to ease inflammation and pain. They include aspirin, ibuprofen, diclofenac and celecoxib.

Topical analgesics. Corticosteroids.

Joint injection

Other drugs. The antidepressant duloxetine and the anti-seizure drug pregabalin. A local steroid injection may be given to reduce inflammation and pain in a joint.

Surgical treatments

damaged joints to restore mobility and relieve pain. Hips and knees are the joints most commonly replaced.

Nonpharmacological treatments

- Exercise.
- Weight loss.
- Adequate sleep.
- Heat and cold therapy.
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Complications

Osteoarthritis becomes dangerous enough to affect in daily activities. And sometimes there are more serious complications. Such as Sleep disruption

Reduced productivity

Weight gain

Anxiety and depression

Possible complications of osteoarthritis include:

- Rapid, complete breakdown of cartilage resulting in loose tissue material in the joint.
- Bone death.
- Stress fractures.
- Bleeding inside the joint.
- Infection in the joint.
- Rupture of the tendons and ligaments around the joint, leading to loss of stability.
- Pinched nerve (in osteoarthritis of the spine).

Prevention

Primary:

Health promotion:

Diet : High protein ,calcium intake ,vitamins & minerals and Omega-3 fatty acids.

Life style : Control your blood sugar, Weight reduction, Be active every day. Specific

protection: Prevention of injuries, When jumping, land with knees bent. Perinatal

screening for congenital hip disease.

Keep your muscles strong.

Health Education:

Physical therapy and exercise programs — Physical therapy and exercise improve flexibility and strengthen the muscles surrounding the joints. People who exercise regularly despite their arthritis will typically have less pain and better function than those who are inactive

Secondary:

Early diagnosis: MRI ,x-ray , Blood tests ,elevated CRP, Vitamin D levels.

Treatment:Nonsteroidal anti-inflammatory drugs ,Topical analgesics,
Corticosteroids.

Tertiary:

Follow up: 1-year follow-up

Prevent complications : Sleep disruption , Reduced productivity, Weight gain,
Anxiety and depression.

Rehabilitation: land- and water-based exercise, strength training, weight management, self-management and education, biomechanical interventions, and physically active lifestyle.

Nursing management

Diagnosis : Acute Pain

Related to :

- Bone deformities
- Joint degeneration
- Muscle spasm
- Physical mobility

Interventions:

- Assess the client's description of pain.
- Assess the client's previous experiences with pain and painrelief.
- Identify factors or activities that seem to precipitate acuteepisodes or aggravate a chronic condition.
- Determine whether the client is reporting all of the pain he orshe is

experiencing.

- Determine the client's emotional reaction to chronic pain.
- Develop pain relief regimen based on the client's identified aggravating and relieving factors.
- Instruct the client to take prescribed analgesics and/or anti-inflammatory medications.

Desired Outcomes

- Client will report satisfactory pain control at a level less than 3 to 4 on a scale of 0 to 10.
- Client will use pharmacological and nonpharmacological pain relief strategies.
- Client will exhibit increased comfort such as baseline levels for HR, BP, respirations, and relaxed muscle tone or body posture.
- Client will engage in desired activities without an increase in pain level.

2-Diagnosis : 1. Inability to move purposefully within the physical environment, including bed mobility, transfers, and ambulation.

Manifested by :- limited ROM - inability to move

Interventions:

- Assist patient for muscle exercises as able or when allowed out of bed; execute abdominal-tightening exercises and knee bends; hop on foot; stand on toes
- Present a safe environment: bed rails up, bed in a down position, important items close by.
- Promote and facilitate early ambulation when possible. Aid with each initial change: dangling legs, sitting in chair, ambulation.
- Let the patient accomplish tasks at his or her own pace. Do not hurry the patient. Encourage independent activity as able and safe.
- Give positive reinforcement during activity. Patients may be unwilling to move or initiate new activity because of fear of falling.
- Provide the patient of rest periods in between activities. Consider energy-saving techniques.
- Give medications as appropriate. Antispasmodic medications may reduce muscle spasms or spasticity that interferes with mobility; analgesics may reduce pain that impedes movement.

Desired Outcomes

- The goals of interventions are to avoid the hazards of immobility, prevent dependent disabilities, and assist the patient in restoring, preserving, or maintaining as

much mobility and functional independence as possible

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